Detectia limbajului ofensiv, urii & misoginiei in texte romanesti

Pipeline unificat 5-clase · TF-IDF baselines · RoBERT

Autori

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NLP hate speech Romanian TF-IDF + BERT

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Cuprins

1. Motivatie
2. Related Work
3. Date
4. Preprocesare
5. Modele
6. Rezultate
7. Perspective
8. Concluzii

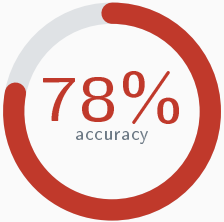
Motivatie

De ce hate speech in romana?

- **Limba cu putine resurse**
  putine corpusuri publice de hate speech
- **Text social zgomotos**
  diacritice lipsa, slang, leetspeak
- **Dezechilibru sever de clase**
  rasism + homofobie $< 1\%$
- **Sens dependent de context**
  insulta vs. citat vs. critica

Capcana acuratetei

Modelul care prezice *mereu* non-offensive:



...dar **nu detecteaza nimic**.

$\Rightarrow$ macro-F1 este metrica corecta.

Contributiile pe scurt

Corpus unificat

news-ro-offense ∪ CoRoSeOf
→ **43,061** exemple, 5 clase

Comparatie controlata

LogReg vs. Calibrated Linear SVM
TF-IDF word+char comun, + RoBERT

Preprocesare adaptata romanei

diacritice, leetspeak, stemming,
emoji, stopwords cu negatii

Ablatie si error analysis

Ce influenteaza macro-F1
+ unde gresesc modelele

Related Work

Peisajul domeniului

- **OffensEval 2019/2020**
  taxonomie ierarhica, multilingv
- **HatEval 2019**
  ura impotriva femeilor si imigrantilor
- **Survey-uri** (Fortuna & Nunes 2018; Schmidt & Wiegand 2017)
- **Modele perspective-aware**
  (Akhtar et al., 2021)

Idee cheie (Akhtar et al., 2021)

Dezacordul intre adnotatori este *semnal*, nu zgomot.

$\Rightarrow$ separam abuzul targetat pe identitate de insulta generica.

Date

Doua surse complementare

Sursa	Randuri	Rol
CoRoSeOf	39,008	sexism / ofensa
news-ro-offense	4,053	5-class hate / offense
Unificat	**43,061**	schema comuna

Schema

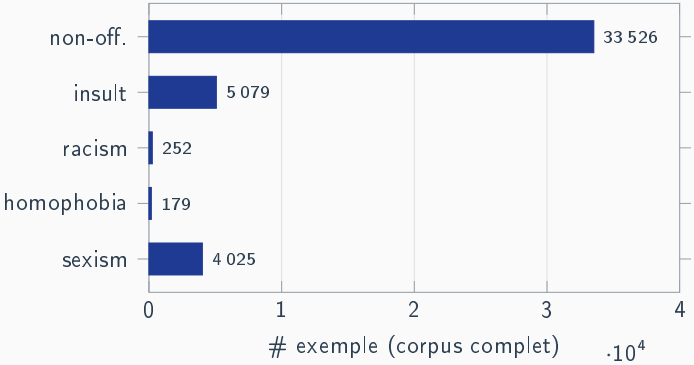
`text` · `label_5` · `offensive_flag` · `sexism_flag` · `target` · `sexism_subtype` · `quality`

Taxonomia cu cinci clase

ID	Clasa	Semnificatie
0	non-off.	text neofensiv
1	insult	insulta generica
2	racism	continut rasist
3	homophobia	continut homofob
4	sexism	sexism / misoginie

Maparea pastreaza subtipurile CoRoSeOf (*direct*, *descriptive*, *reporting*) in `sexism_subtype`.

Distributia: o coada lunga de ura



Long tail

$\approx 1\%$

Rasism + homofobie din intregul corpus.

Non-offensive domina; clasele de ura targetata sunt extrem de rare.

Subset experimental stratificat

Compozitie

- **800** non-off. (seed=42)
- **200** insult **200** sexism
- **toate** racism (252)
- **toate** homophobia (179)

1,631

randuri total — 800 non-off / 831 ofensive

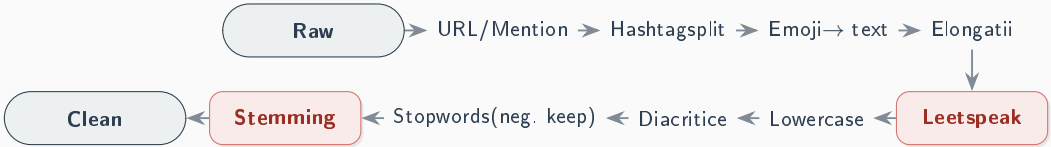
Split stratificat 70 / 15 / 15

Split	N	0	1	2	3	4
Train	1,141	560	140	176	125	140
Val	245	120	30	38	27	30
Test	245	120	30	38	27	30

Stratificare per-clasa $\Rightarrow$ fiecare clasa apare in fiecare partitie.

Preprocesare

Preprocesare adaptata romanei



Exemplu pas cu pas

Brut	*@user esti pr0asta!! #romani*
mention/hashtag	`<USER> esti pr0asta romani`
leetspeak	`<USER> esti proasta romani`
lowercase	`<user> esti proasta romani`
stopwords	`<user> proasta romani`

130k → 12.7k

features dupa stemming

Modele

Spatiu de features comun

TF-IDF: word + character

word $(1,2)$-grams $\oplus$ char $(3,5)$-grams

`min_df=3, max_df=0.9, sublinear_tf=True`

$\approx$ **20,000** features (subset demo)

Acelasi vectorizer, fit doar pe *train* $\Rightarrow$ diferentele vin din clasificator.

Trei clasificatori, un singur feature space

LogReg

softmax posterior

$C \in \{10^{-3}, \ldots, 10^{2}\}$

best $C = 0.3$

`class_weight=balanced`

stem=True

Linear SVM

max-margin hyperplane

$C \in \{0.1, \ldots, 5.0\}$

best $C = 0.5$

+ **Platt calibration** (5-fold)

stem=False

RoBERT

Romanian BERT

AdamW, lr $= 2 \cdot 10^{-5}$

batch 16, 5 epochs

classifier head $\rightarrow$ 5 clase

baseline contextual

5-fold `StratifiedKFold` CV — evita fold-uri fara homophobia (doar 125 ex. train).

Rezultate

Cifrele principale

Model	**Acc.**	**F1$_{\text{macro}}$**	F1$_{\text{ins.}}$	F1$_{\text{rac.}}$	F1$_{\text{hom.}}$	F1$_{\text{sex.}}$
LogReg	0.751	0.698	0.517	0.784	0.667	0.690
Linear SVM	**0.788**	**0.720**	0.490	**0.833**	0.640	**0.772**

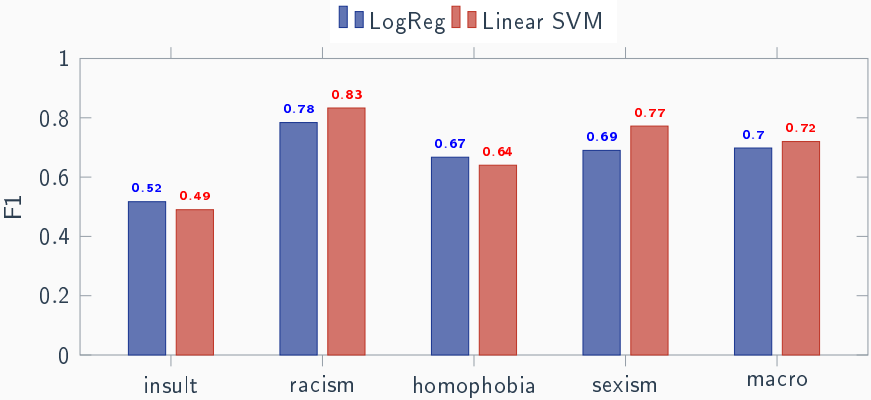
+0.022

macro-F1, SVM peste LogReg

+0.082

F1 pe *sexism*, SVM peste LogReg

Macro-F1 pe clase



SVM castiga pe clasele minoritare; *insult* ramane cea mai grea.

Ablatie — ce a contat de fapt?

Modificare (LogReg)	**Macro-F1**	**Δ**
Baseline	0.671	—
+ Stemming (130k → 12.7k feats.)	0.691	+0.020
+ `min_df=3, max_df=0.9`	0.698	+0.007
+ char $(3,5)$ vs. $(2,4)/(3,6)$	0.698	cea mai buna
Final	**0.698**	

Pentru SVM: vectorizer comun word+char + Platt calibration + `dual=auto` ⇒ 0.645 → **0.720**.

Unde gresesc inca modelele

- insult $\leftrightarrow$ sexism
  granita dependenta de adnotare
- **homophobia** cea mai grea
  doar 125 exemple train
- **Mentiune neutra $\rightarrow$ non-off.**
  lipseste contextul pragmatic

Pattern

Bottleneck-ul nu mai este *modelul* — ci *datele*.

Clasele rare au < 40 ex. pe dev; o singura greseala muta F1 cu cateva puncte.

Perspective

Future work

- **Extinderea claselor rare**
  adnotare targetata pentru racism si homophobia
- **Cross-lingual transfer**
  folosirea corpusurilor de hate speech engleze/italiene
- **Modele perspective-aware** (Akhtar et al., 2021)
  label-uri multi-annotator, ensemble pe dezacord
- **Scenariu de deployment**
  flag-and-defer catre moderator inainte de publicare

Limitari

- **Clase rare**
  179 ex. homophobia in total
- **Domain mismatch**
  stiri vs. text social
- **Doar romana**
  retraining necesar pentru transfer
- **Limitarile TF-IDF**
  rateaza sarcasm, stereotip implicit
- **Subiectivitate la adnotare**
  un singur label ascunde dezacordul

Declaratie etica

Setul de date contine **limbaj ofensiv si potential daunator**.

Modelele propuse sunt destinate doar pentru **cercetare si asistenta la moderare**.

Orice deployment in productie necesita revizuire umana pentru a evita decizii automate incorecte sau injuste.

Concluzii

Trei idei de retinut

1. **Unificarea conteaza.**
   Doua corpusuri eterogene → o taxonomie utila cu 5 clase.
2. **SVM calibrat bate LogReg.**
   0.720 vs. 0.698 macro-F1 pe acelasi feature space.
3. **Datele, nu modelul, sunt plafonul.**
   Clasele minoritare misca macro-F1 — mai multe exemple.

Multumim!

Intrebari?

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